Program 6: MAVEN TO GRADLE

STEP 1:Create a new maven project in Devops by using the command

mvn archetype:generate -DgroupId=com.example -DartifactId=MyMavenToGradle -DarchetypeArtifactiId=maven-archetype-quickstart -DinteractiveMode=false

```
brijesh@Ubuntu:~/Devops$ mvn archetype:generate -DgroupId=com.example -DartifactId=MyMavenToGradle -DarchetypeArtifactiId=ma
[INFO] Scanning for projects...
[INFO]
[INFO] ------------------< org.apache.maven:standalone-pom >-------------------
[INFO] Building Maven Stub Project (No POM) 1
[INFO] --------------------------------[ pom ]---------------------------------
[INFO]
[INFO] >>> maven-archetype-plugin:3.4.1:generate (default-cli) > generate-sources @ standalone-pom >>>
[INFO]
[INFO] <<< maven-archetype-plugin:3.4.1:generate (default-cli) < generate-sources @ standalone-pom <<<
[INFO]
[INFO]
[INFO] --- maven-archetype-plugin:3.4.1:generate (default-cli) @ standalone-pom ---
[INFO] Generating project in Batch mode
Downloading from central: https://repo.maven.apache.org/maven2/archetype-catalog.xml
Downloaded from central: https://repo.maven.apache.org/maven2/archetype-catalog.xml (18 MB at 6.0 MB/s)
[INFO] No archetype defined. Using maven-archetype-quickstart (org.apache.maven.archetypes:maven-archetype-quickstart:1.0)
[INFO] ----------------------------------------------------------------------------
[INFO] Using following parameters for creating project from Old (1.x) Archetype: maven-archetype-quickstart:1.0
[INFO] ----------------------------------------------------------------------------
[INFO] Parameter: basedir, Value: /home/brijesh/Devops
[INFO] Parameter: package, Value: com.example
[INFO] Parameter: groupId, Value: com.example
[INFO] Parameter: artifactId, Value: MyMavenToGradle
[INFO] Parameter: packageName, Value: com.example
[INFO] Parameter: version, Value: 1.0-SNAPSHOT
[INFO] project created from Old (1.x) Archetype in dir: /home/brijesh/Devops/MyMavenToGradle
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  8.727 s
[INFO] Finished at: 2026-03-25T08:57:28+05:30
[INFO] ------------------------------------------------------------------------
brijesh@Ubuntu:~/Devops$
```

STEP 2: EDIT pom.xml

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
http://maven.apache.org/maven-v4_0_0.xsd">
<modelVersion>4.0.0</modelVersion>
<groupId>com.example1</groupId>
<artifactId>MyMavenToGradle</artifactId>
<packaging>jar</packaging>
<version>1.0-SNAPSHOT</version>
<name>MyMavenToGradle</name>
<url>http://maven.apache.org</url>
<!--  Properties: Customize Java version or plugin versions  -->
<properties>
<maven.compiler.source>11</maven.compiler.source>
<maven.compiler.target>11</maven.compiler.target>
```

```
</properties>
<dependencies>
<dependency>
<groupId>junit</groupId>
<artifactId>junit</artifactId>
<version>3.8.1</version>
<scope>test</scope>
</dependency>
</dependencies>
<!--  Build: Configuring plugins and build settings  -->
<build>
<plugins>
<!--  Example: Maven Compiler Plugin to compile Java code  -->
<plugin>
<groupId>org.apache.maven.plugins</groupId>
<artifactId>maven-compiler-plugin</artifactId>
<version>3.8.1</version>
<configuration>
<source>1.6</source>
<target>1.6</target>
</configuration>
</plugin>
<!--  Example: Maven Surefire Plugin to run tests  -->
<plugin>
<groupId>org.apache.maven.plugins</groupId>
<artifactId>maven-surefire-plugin</artifactId>
<version>2.22.2</version>
</plugin>
<plugin>
<groupId>org.apache.maven.plugins</groupId>
<artifactId>maven-jar-plugin</artifactId>
<version>3.1.0</version>
<configuration>
<archive>
<manifestEntries>
<Main-Class>com.example1.App</Main-Class>
</manifestEntries>
</archive>
</configuration>
</plugin>
</plugins>
</build>
</project>
```

STEP 3: Execute the command mvn clean install

```
brijesh@Ubuntu:~/Devops/MyMavenToGradle$ mvn clean install
[INFO] Scanning for projects...
[INFO]
[INFO] --------------------< com.example1:MyMavenToGradle >--------------------
[INFO] Building MyMavenToGradle 1.0-SNAPSHOT
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- maven-clean-plugin:2.5:clean (default-clean) @ MyMavenToGradle ---
[INFO]
[INFO] --- maven-resources-plugin:2.6:resources (default-resources) @ MyMavenToGradle ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /home/brijesh/Devops/MyMavenToGradle/src/main/resources
[INFO]
[INFO] --- maven-compiler-plugin:3.8.1:compile (default-compile) @ MyMavenToGradle ---
[INFO] Changes detected - recompiling the module!
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file to /home/brijesh/Devops/MyMavenToGradle/target/classes
[INFO]
[INFO] --- maven-resources-plugin:2.6:testResources (default-testResources) @ MyMavenToGradle ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /home/brijesh/Devops/MyMavenToGradle/src/test/resources
[INFO]
[INFO] --- maven-compiler-plugin:3.8.1:testCompile (default-testCompile) @ MyMavenToGradle ---
[INFO] Changes detected - recompiling the module!
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file to /home/brijesh/Devops/MyMavenToGradle/target/test-classes
[INFO]
[INFO] --- maven-surefire-plugin:2.22.2:test (default-test) @ MyMavenToGradle ---
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running com.example.AppTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.014 s - in com.example.AppTest
[INFO]
```

STEP 4: Now execute the command “gradle init –type pom.xml”, we get a error for this

```
brijesh@Ubuntu:~/Devops/MyMavenToGradle$ gradle init --type pom.xml

FAILURE: Build failed with an exception.

* What went wrong:
Execution failed for task ':init'.
> The requested build setup type 'pom.xml' is not supported. Supported types: 'basic', 'groovy-applicatio

* Try:
Run with --stacktrace option to get the stack trace. Run with --info or --debug option to get more log ou

* Get more help at https://help.gradle.org

BUILD FAILED in 6s
2 actionable tasks: 2 executed
brijesh@Ubuntu:~/Devops/MyMavenToGradle$
```

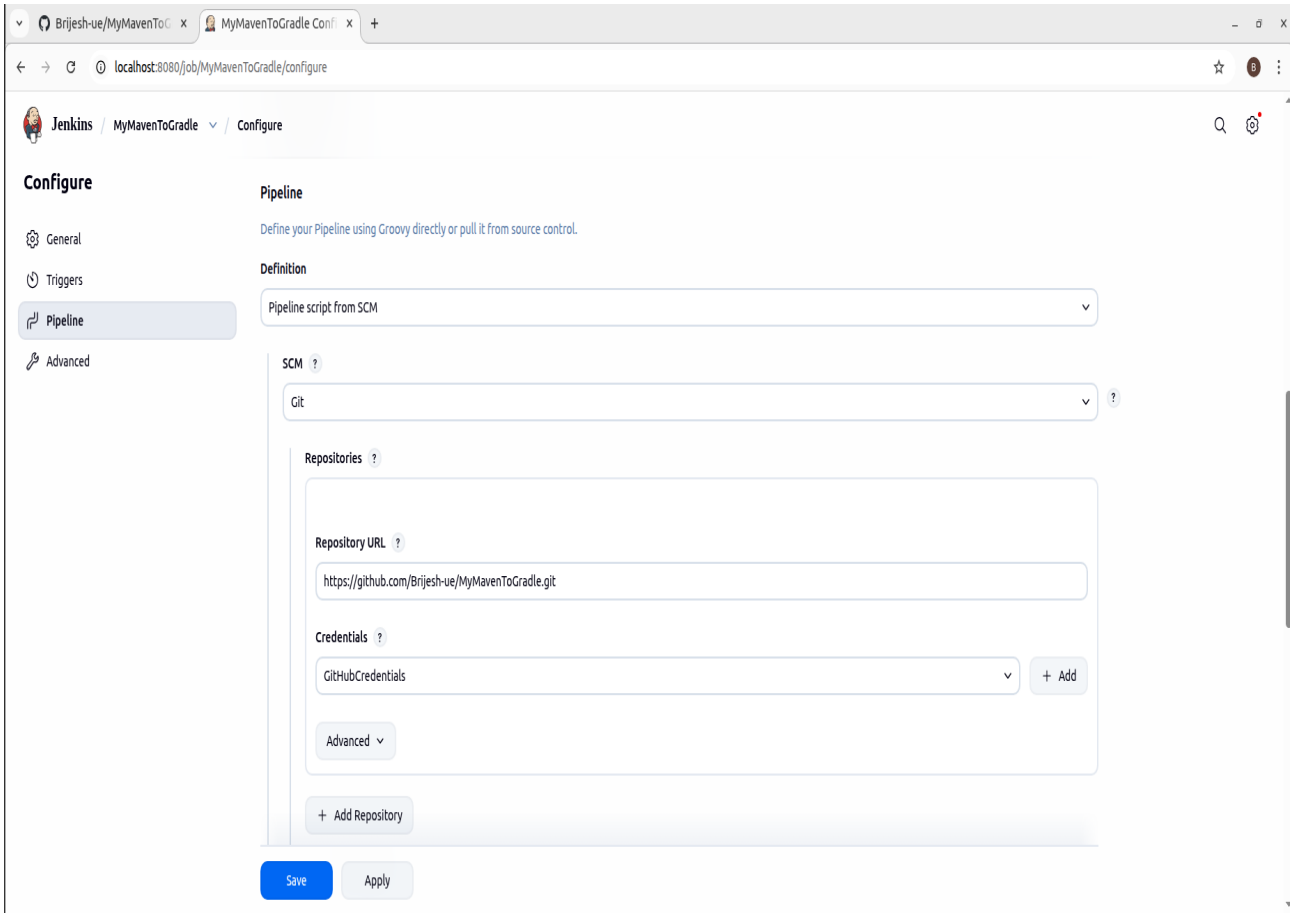
STEP 5: Now execute the command “gradle init –type pom”, we get build success

```
brijesh@Ubuntu:~/Devops/MyMavenToGradle$ gradle init --type pom

> Task :init
Maven to Gradle conversion is an incubating feature.


BUILD SUCCESSFUL in 5s
2 actionable tasks: 1 executed, 1 up-to-date
brijesh@Ubuntu:~/Devops/MyMavenToGradle$
```

MavenToGradle Pipeline:

Step 1: Create a New Pipeline Job
Go to Dashboard → + New Item.
Enter name, select Pipeline, and click OK.

Step 2:Configure Pipeline from SCM
Scroll down and select Pipeline script from SCM.
SCM: Git Enter repository URL and select GitHub credentials.

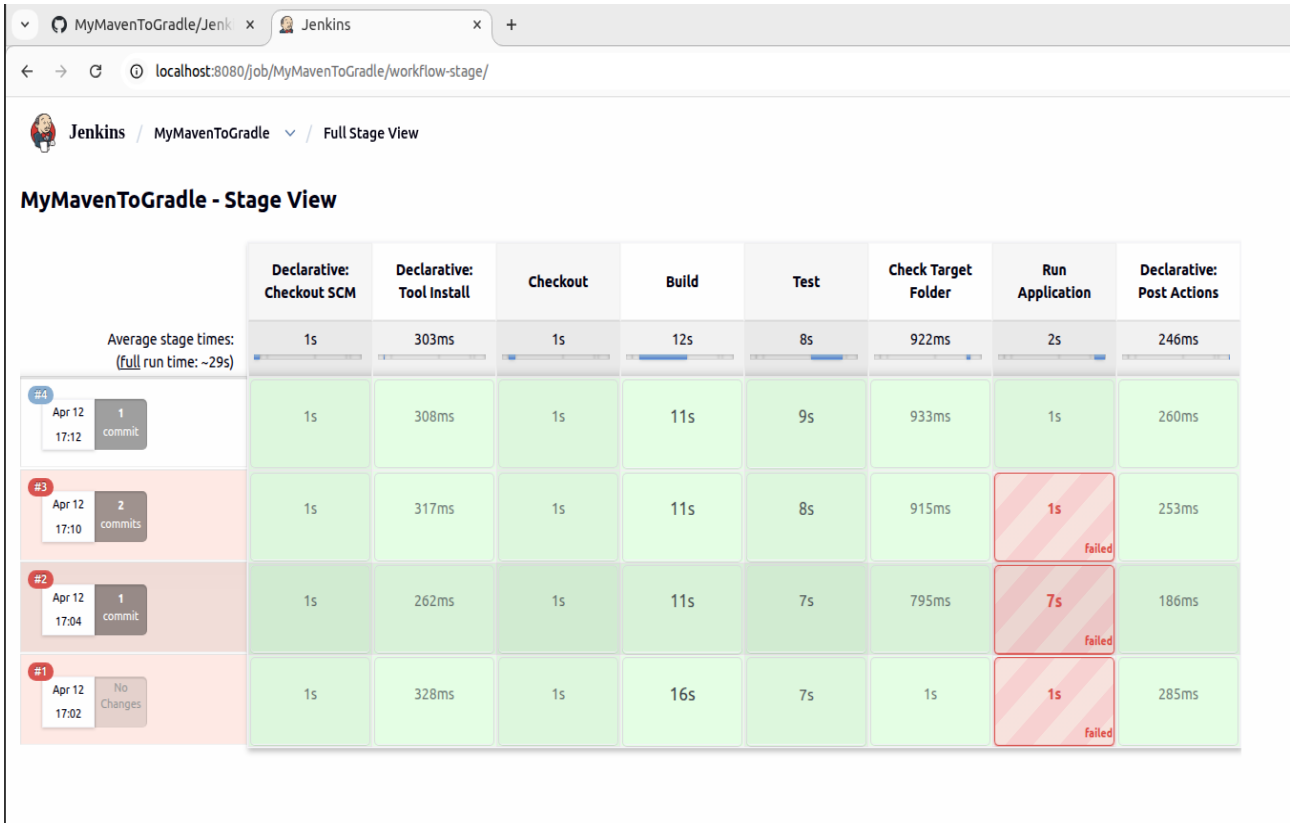


Step 3: Create Jenkinsfile

```
pipeline {
    agent any

    tools {
        maven 'Maven'
    }

    stages {

        stage('Checkout') {
            steps {
                git branch: 'main', url:
'https://github.com/Brijesh-ue/MyMavenToGradle.git'
            }
        }

        stage('Build') {
            steps {
                sh 'mvn clean package'
            }
        }

        stage('Test') {
            steps {
                sh 'mvn test'
            }
        }

        stage('Check Target Folder') {
            steps {
                sh 'ls -l target'
            }
        }

        stage('Run Application') {
            steps {
                sh 'java -jar target/MyMavenToGradle-1.0-SNAPSHOT.jar'
```

```
            }
        }

    }

    post {
        success {
            echo 'Build and execution successful!'
        }
        failure {
            echo 'Build failed!'
        }
    }
}
```

Step 4: Build the Pipeline

Go to Jenkins and click Build Now.

Monitor Build Stages Use Stage View and Full Stage View to monitor execution.